Set 3:

1. **Non-Repudiation:** User cannot deny any action that they have done on the software, must track user action
   Violation: Students submitting assignment late on elms, but elms cannot differentiate between late submission and normal submission

   Authentication: Verify the identification of a user
   Violation: You can login into a system using your username as password

2. **Dictionary Attack:** Hackers try out most common passwords, or use brute force to try all password combinations to hack an account
   Safety: Use strong passwords with symbols, number, capital and small letters, long length / Change your password regularly / Do not use common passwords, or passwords that has your personal information inside

   Trojan Horse: An application that looks useful outside, but is malicious and harmful inside
   Safety: Download applications from authentic websites or app stores / Use antivirus to protect yourself from harmful software

Set 4:

1. **Integrity:** Data can be edited by only authorised person
   Violation: Students being able to edit grades due to a glitch in ucam

   Confidentiality: Data can be viewed by only authorised person
   Violation: You can see cgpa of other students in ucam

2. **SQL Injection:** Attack the database of a system by inserting harmful query through input fields or other means
   Safety: Use SQL Parameters to check input parameters properly / Length checking of input fields / Using high-level languages such as Java

   Man-In-The-Middle: Hackers intercept the network packets and steal data from the packets
   Safety: Use encryption so that an attacker cannot figure out the data even if they intercept the packets / Do authentication on both the client and server side